Photodetectors

The following subjects will be covered

- Photodetector Types.
- General Characteristics of Photoconductors.
- Modes of Operation of Photodiodes.
- PN, PIN and APD Devices.
- Photodiode Circuit Analysis.

Photodetectors

Type (1): Photo-emissive

Light is absorbed, electron-hole pair generated and ejected out of material as current.

Type (2): Photo-conductive

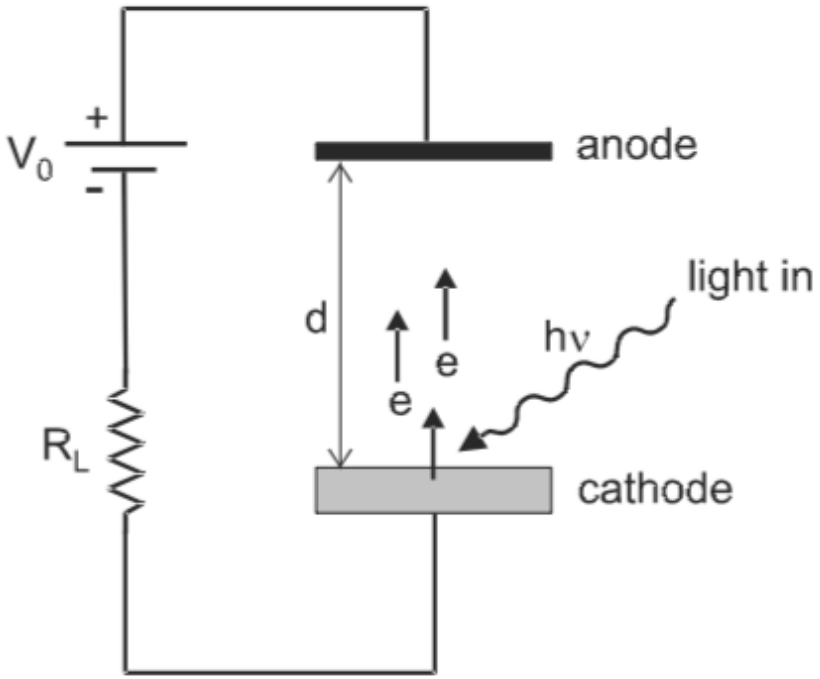
Light is absorbed, electron-hole pair generated and the material conductivity is changed.

Examples:

- Photoconductors: A piece of semiconductor the conductance of which is controlled by light (photoresistors).
- Photodiodes: A pn junction-based device that converts light energy to electron-hole pairs. The ehp are saved and used for operation. (PIN diodes, APDs, Solar cells, etc.)

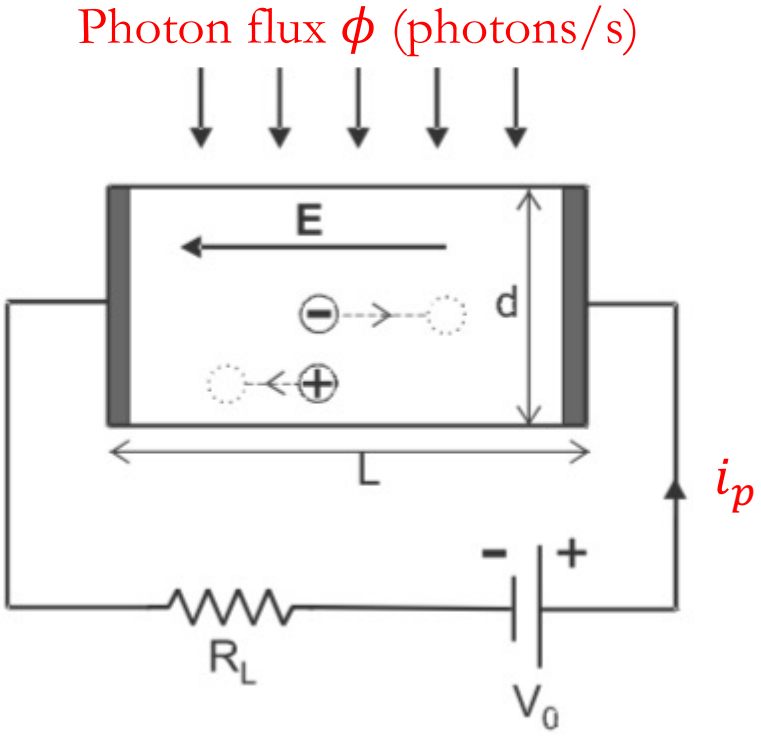
Photo-emissive Devices

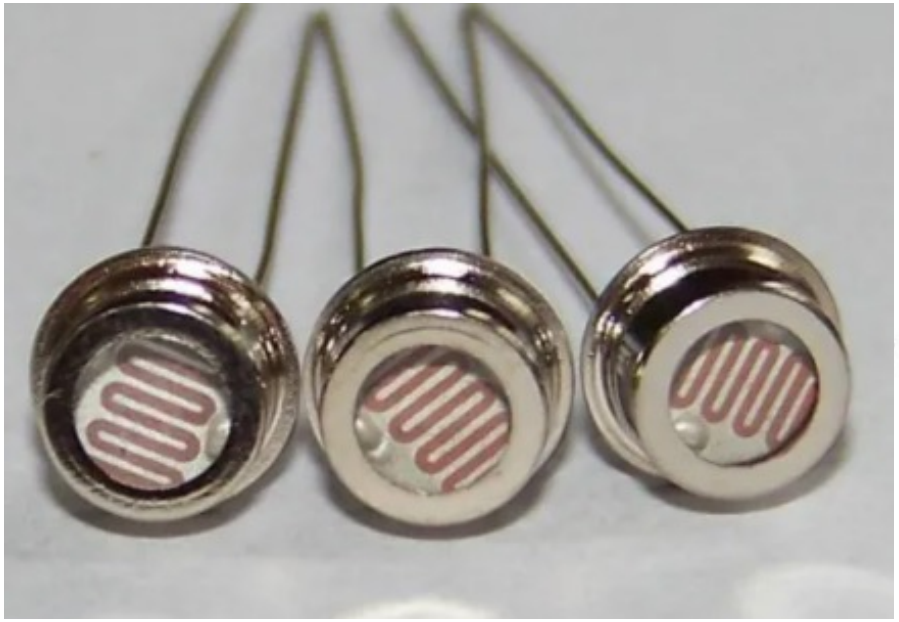
The basic structure of a photo-emissive device is a photoelectric-effect set-up.



Photoconductive Devices

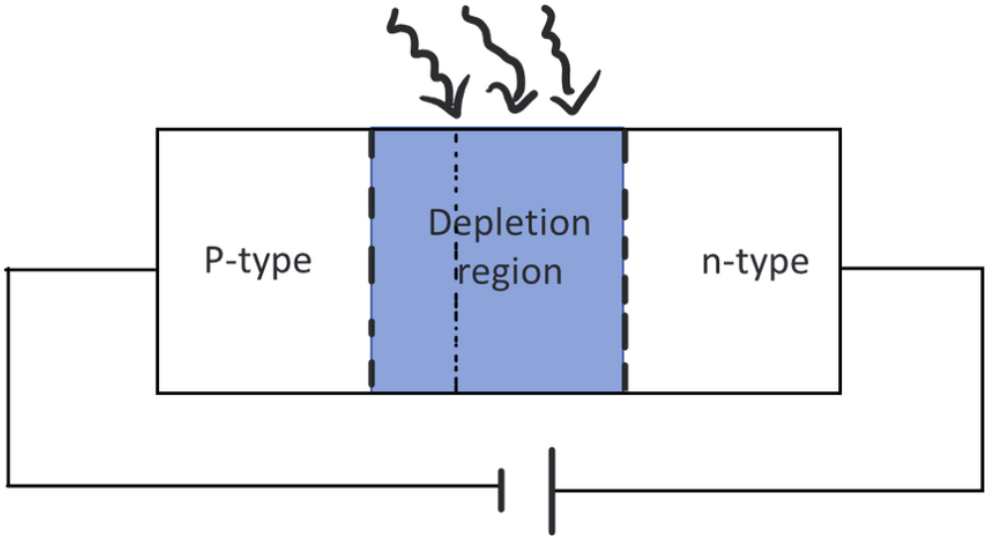
The basic structure of a photoconductive device is a semiconductor material connected across a battery.





Photodiodes

The basic structure of a photodiode device is a pn junction.



Photoconductor Characteristics

Quantum efficiency. The ratio of optical current ehps to incident photon flux.

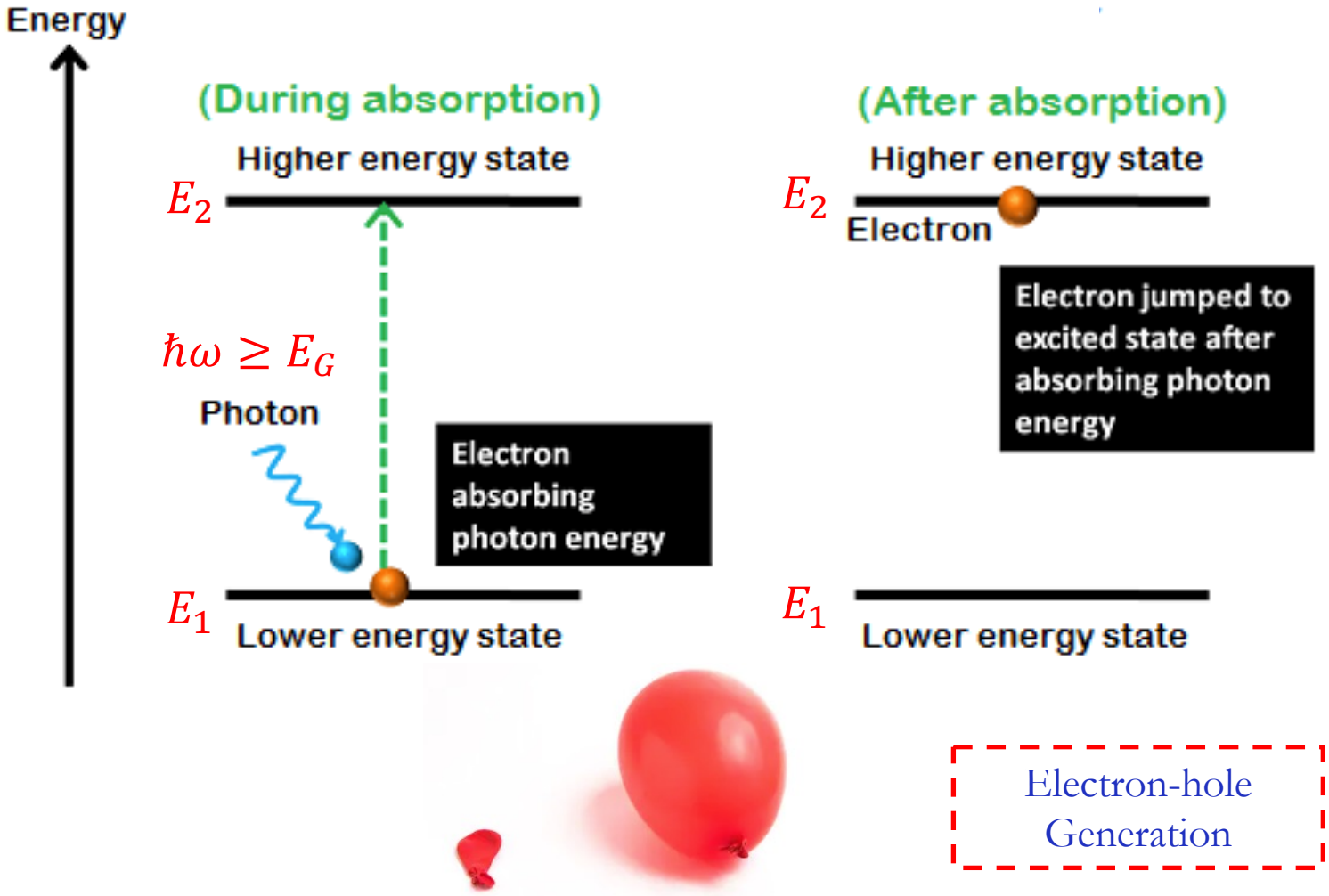
Responsivity. The ratio of optical current to incident optical power.

Impulse response. The rise time of the output current or the output bandwidth.

Noise power. The amount of current created by sources other than light.

Photon Generation and Absorption

Absorption

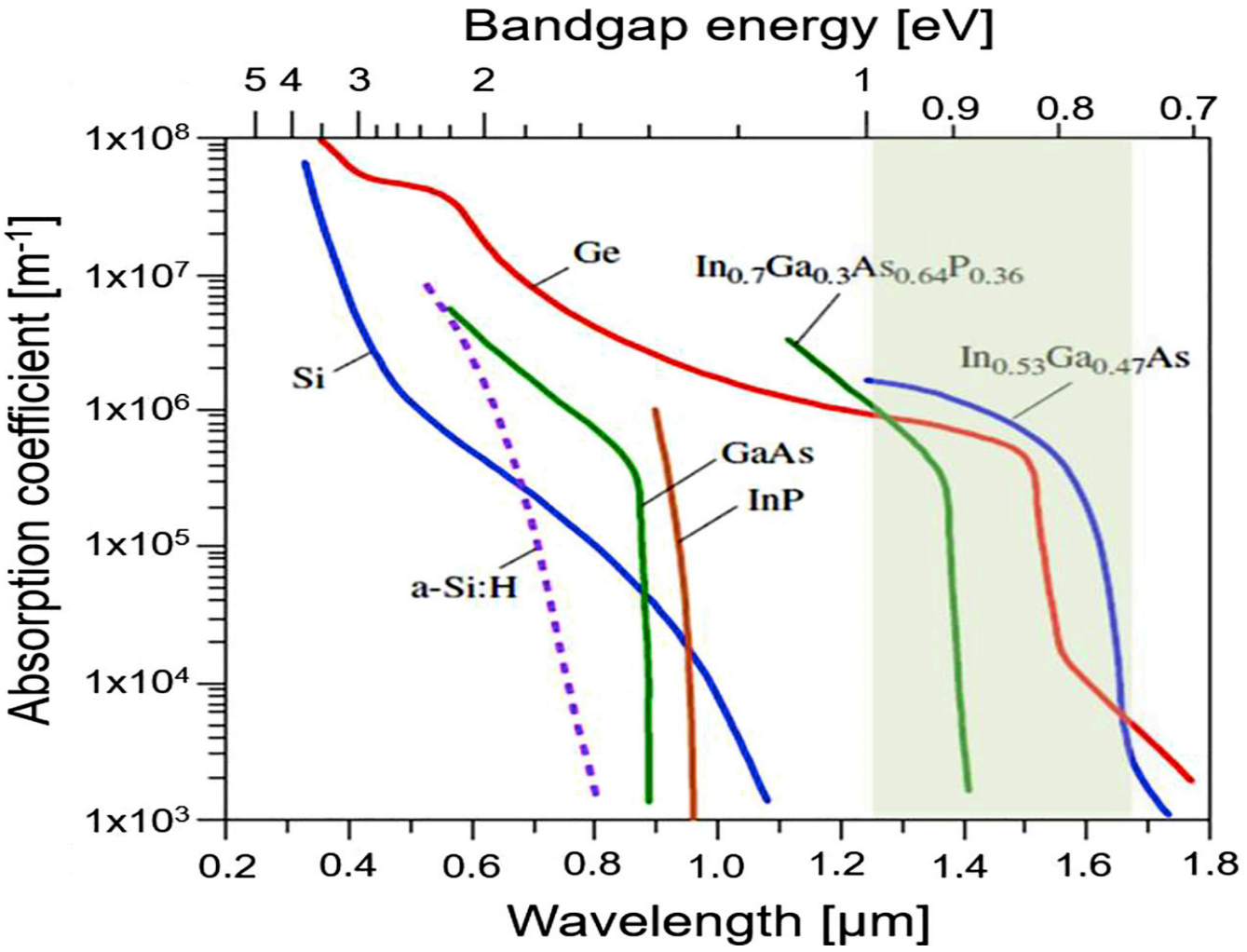


Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

Absorption Curves

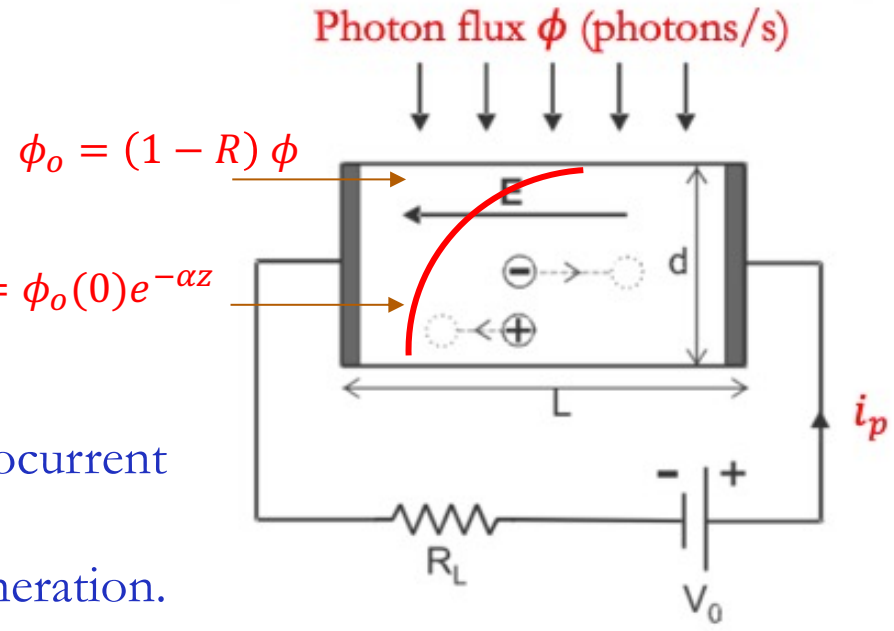
Absorption coefficient of common semiconductor materials.



Photoconductor Characteristics
Efficiency

Quantum efficiency. The ratio of created ehp to number of incident photons.

$$\eta = \frac{\text{ehp flux making the optical current}}{\text{Incident photon flux } (\phi)}$$



Absorbed photon flux:

$$(1-R)\left(1-e^{-\alpha d}\right)\phi$$

Not all absorbed photons will make photocurrent due to:
(1) ehp recombination (2) phonon generation.
Introduce the factor $0 \le \xi \le 1$.

Current-making photon flux = $(1-R)\left(1-e^{-\alpha d}\right)\xi\,\phi$

$$\eta = (1-R)\left(1-e^{-\alpha d}\right)\xi$$

The engineering task is to maximize the efficiency.

Photoconductor Characteristics
Efficiency

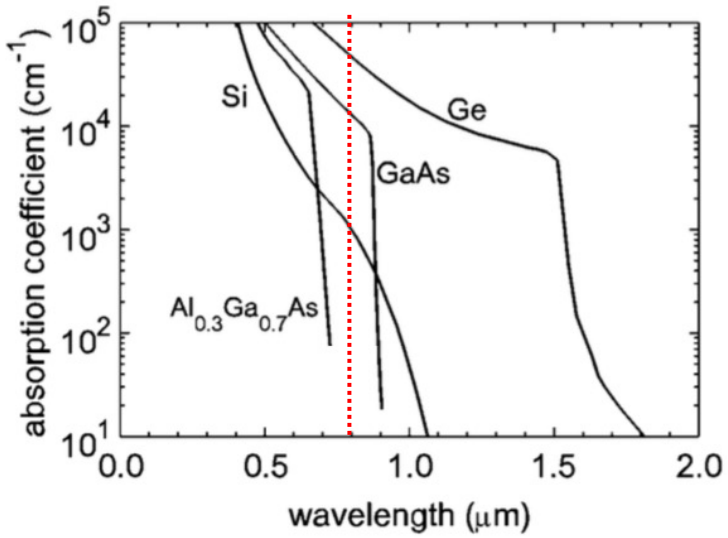
Quantum efficiency. The ratio of created ehp to number of incident photons.

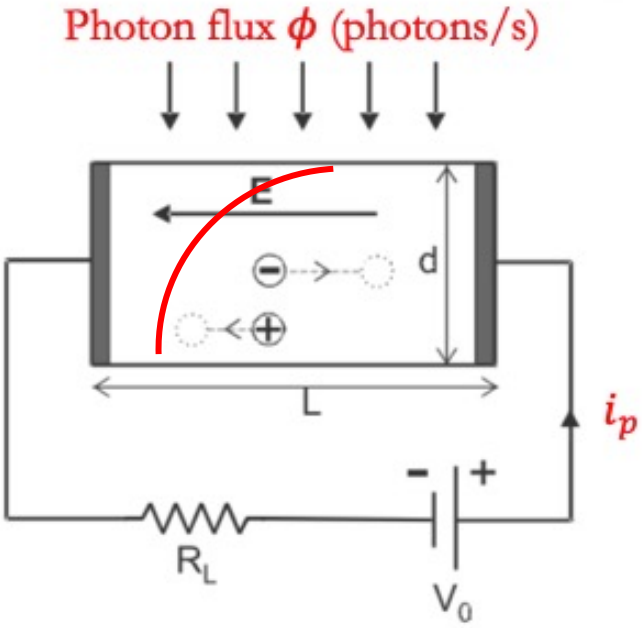
$$\eta = (1 - R)\left(1 - e^{-\alpha d}\right)\xi$$

ξ : Apply bias and clean fabrication process.

R : Use anti-reflection coating (small bandwidth)

αd : Choose best material and thickness.





Example: For 95% absorption at 0.8 μm, find material and thickness.

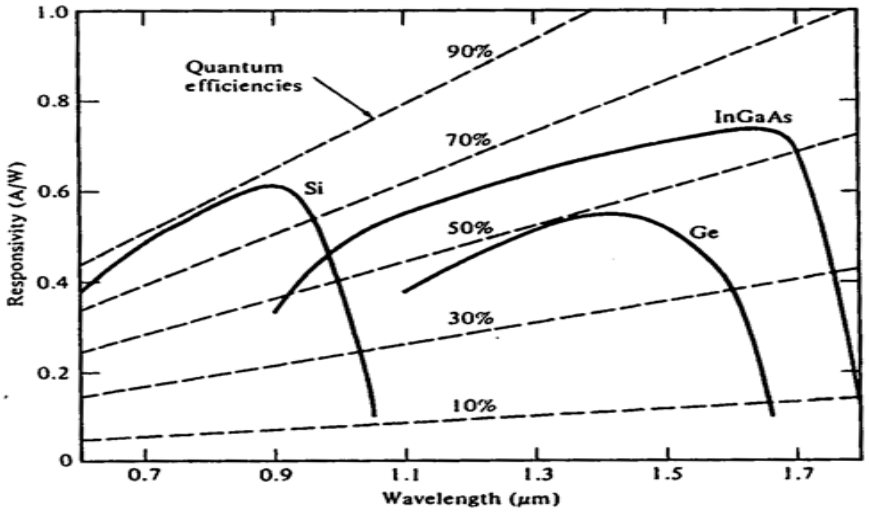
$$1 - e^{-\alpha d} = 0.95$$

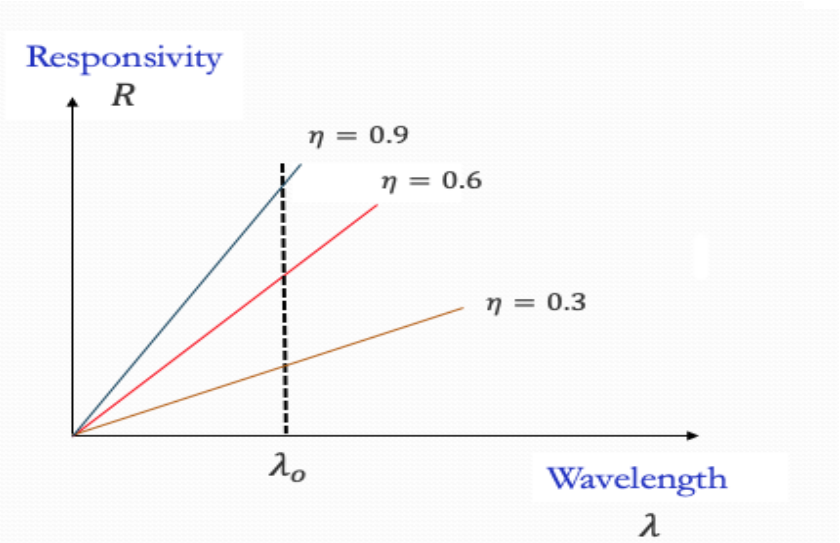
Photoconductor Characteristics
Responsivity

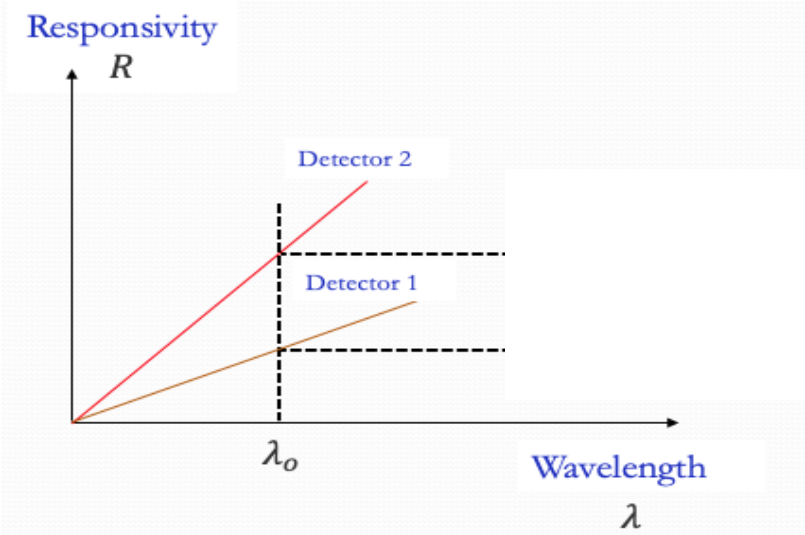
Responsivity. The ratio of optical current to incident optical power.

$$R = \frac{i_p}{P_o} = \eta \frac{e}{\hbar\omega} \qquad \text{(A/W)}$$

$$R = \frac{\eta e}{\hbar 2\pi c}\lambda$$

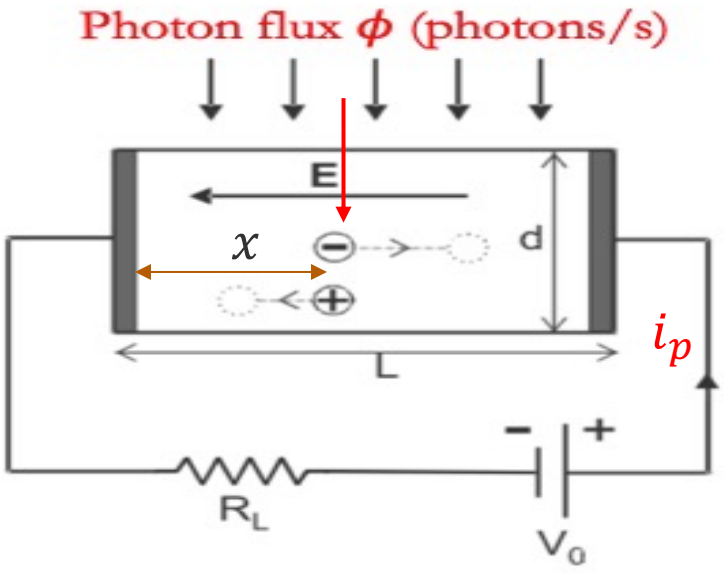


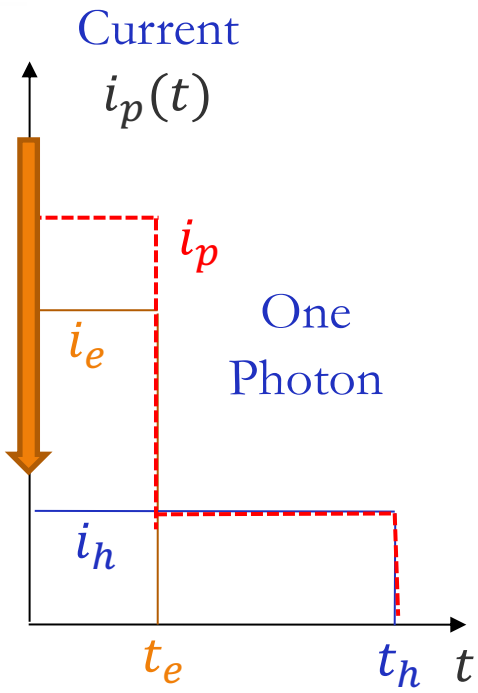


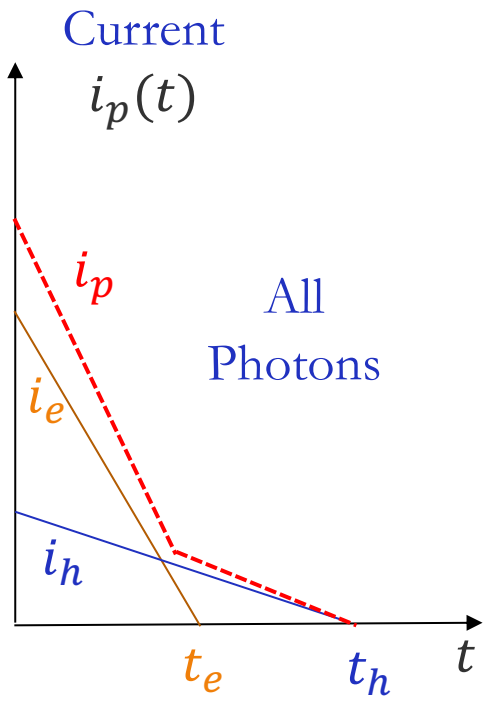


Photoconductor Characteristics
Impulse Response

The impulse response is composed of transit time and RC delay.







$$i_e(t) = e\frac{v_e}{l}$$

$$i_h(t) = e\frac{v_h}{l}$$

$$i_p(t) = \frac{e}{l}(v_e + v_h)$$

$$t_e = \frac{L - x}{v_e}$$

$$t_h = \frac{x}{v_h}$$

Example. GaAs material with L= 1 cm, v_e=10^7 cm/s. i_e=1.6 pA.

Photoconductor Characteristics
Impulse Response

The impulse response is composed of transit time and RC delay.

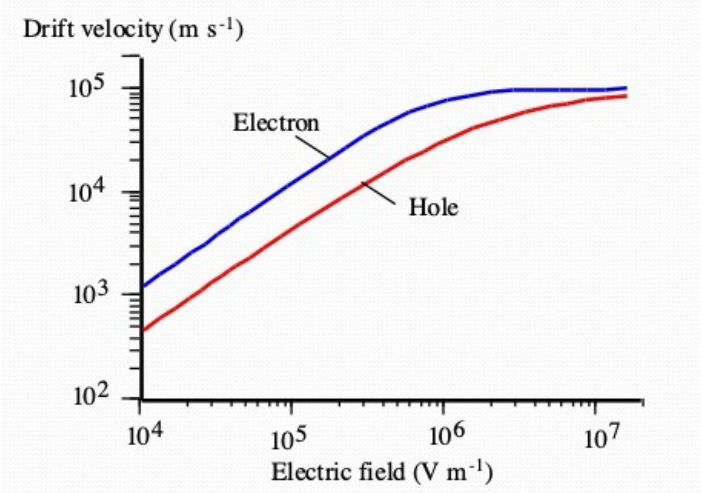
$$i_e(t) = e\frac{v_e}{L}$$

$$i_h(t) = e\frac{v_h}{L}$$

$$i_p(t) = \frac{e}{L}(v_e + v_h)$$

$$v_e = \mu_n E \qquad v_h = \mu_p E$$

cm²/Vs	μ_n	μ_p
Si	1500	450
Ge	3900	1900
InP	4600	150
InGaAs	14000	400



To improve the response:

- High mobility material.
- Small L.
- Drift velocity point.
- Help holes ($t_e = t_h$).

Cut-off Wavelength

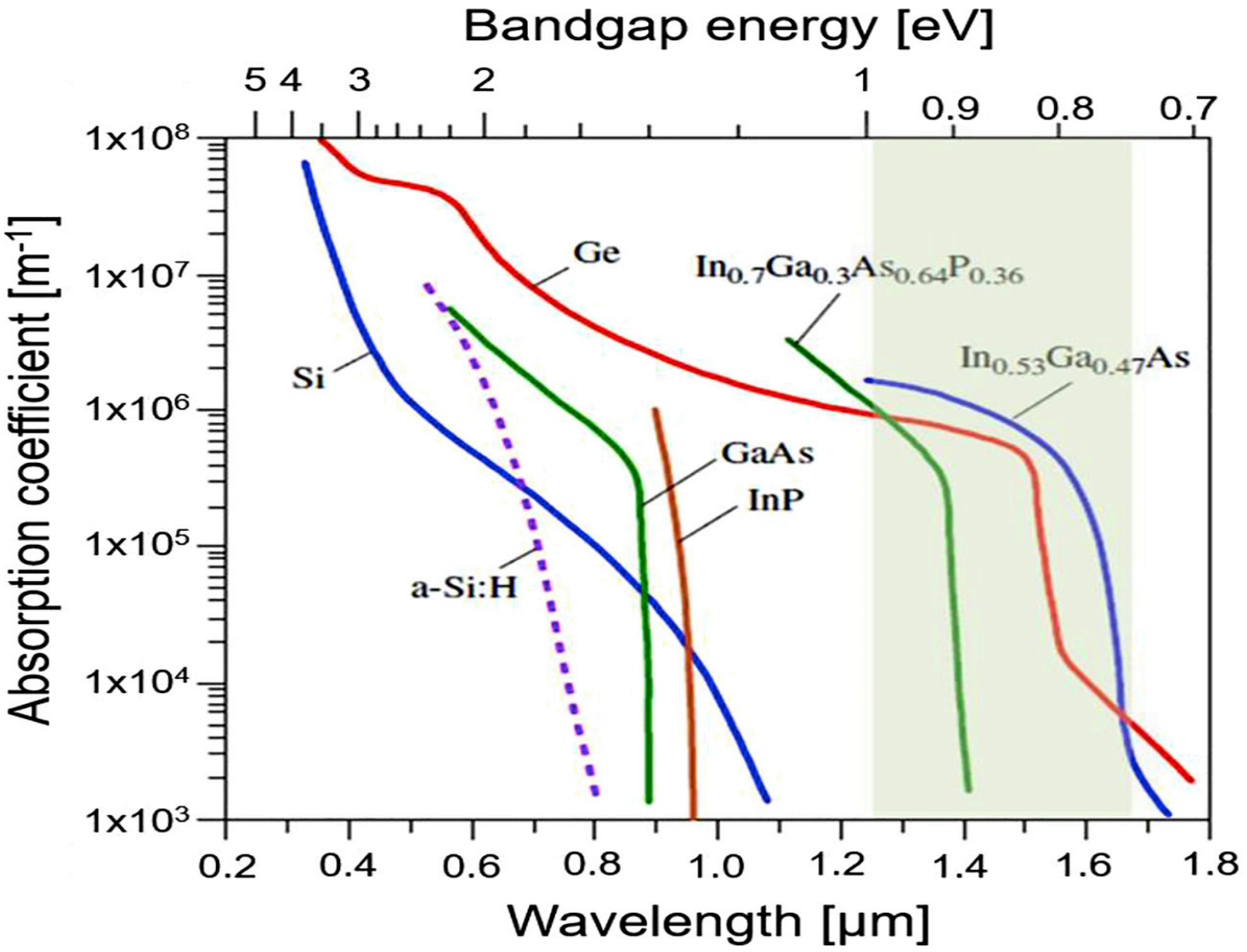
$$\hbar\omega \geq E_g$$

$$\lambda_{co} = \frac{hc}{E_g} = \frac{1.24}{E_g}$$

	E_g(eV)	λ_{co}(μm)
Si	1.12	1.11
Ge	0.66	1.87
InP	1.35	0.91
InGaAs	0.75	1.65

Absorption Curves

Absorption coefficient of common semiconductor materials.

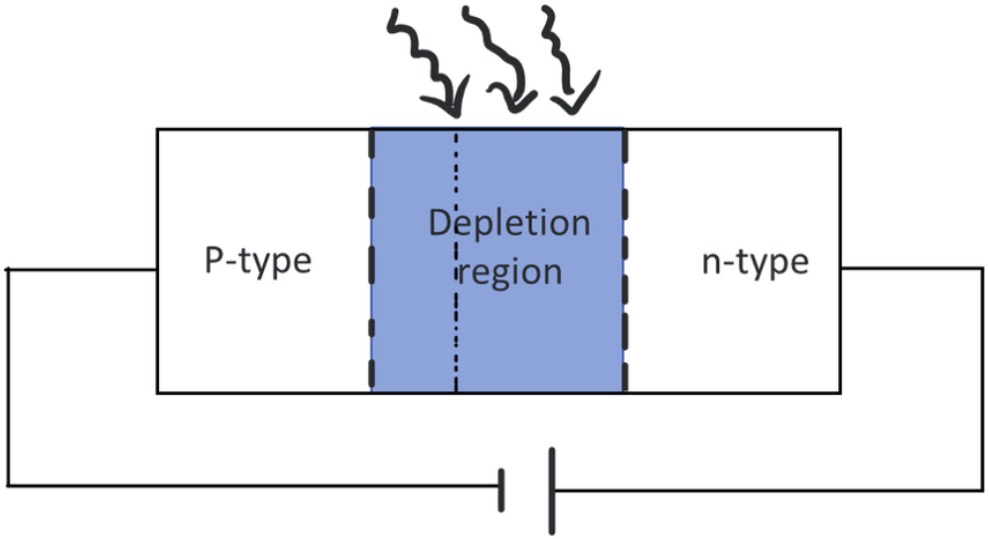


Photoconductor Characteristics: Datasheet

Parameter	Symbol	Unit	Si	Ge	InGaAs
Wavelength range	λ	nm	400-1100	800-1650	1100-1700
Responsivity	R	A/W	0.4-0.6	0.4 –0.5	0.75 – 0.95
Dark current	I_D	nA	1-10	50-500	0.5-2.0
Rise time	τ_r	ns	0.5-1	0.1-0.5	0.05-0.5
Bandwidth	B	GHz	0.3-0.7	0.5-3	1-2
Bias voltage	V_B	V	5	5-10	5

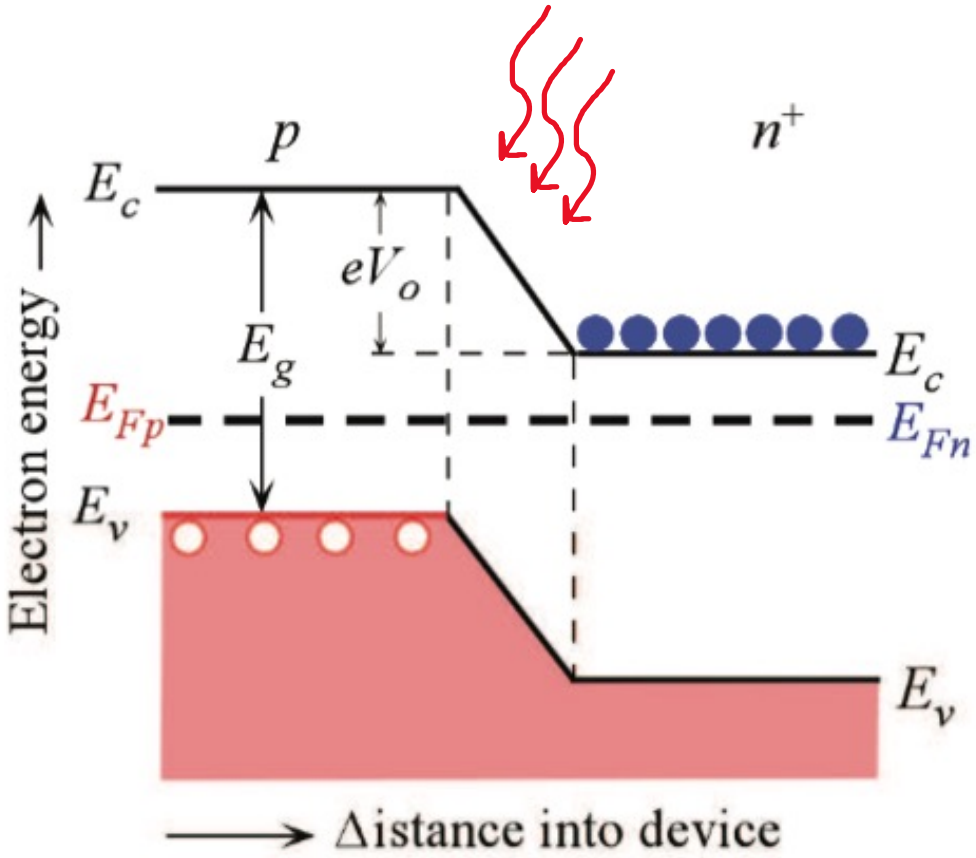
Photodiodes

The basic structure of a photodiode device is a pn junction.



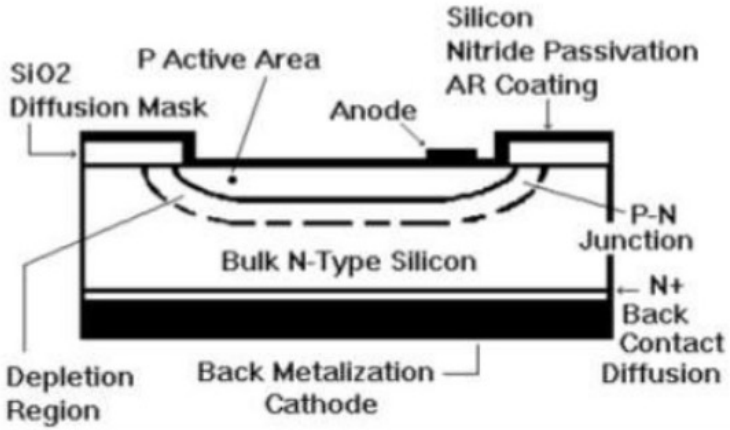
Photodiodes

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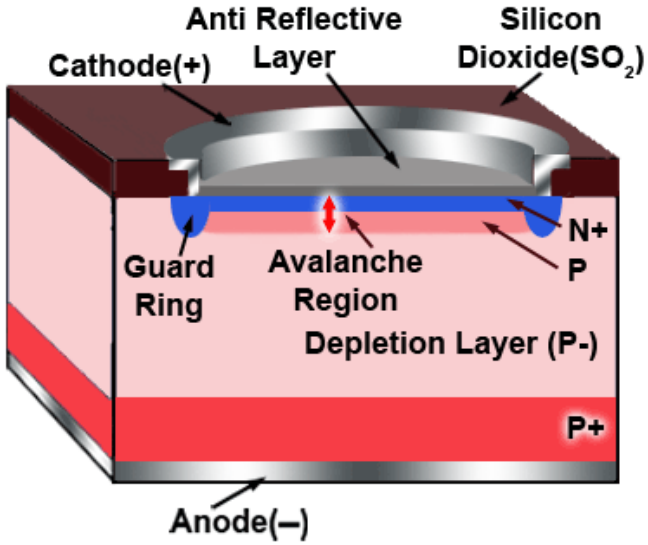


Photodiodes Types and Designs

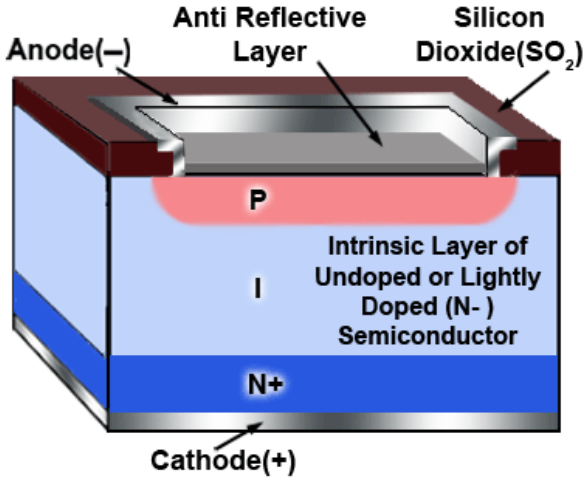
Basic PN photodiode.



Avalanche photodiode (APD).



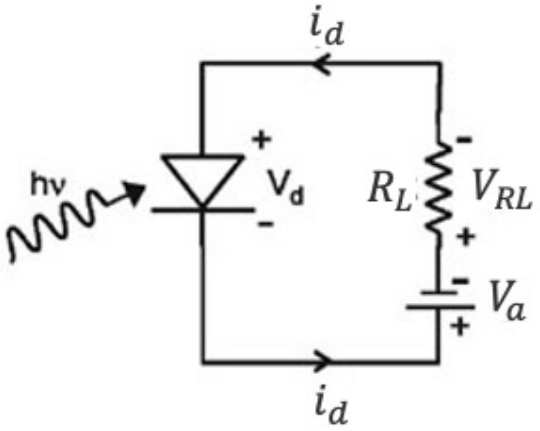
PIN photodiode.



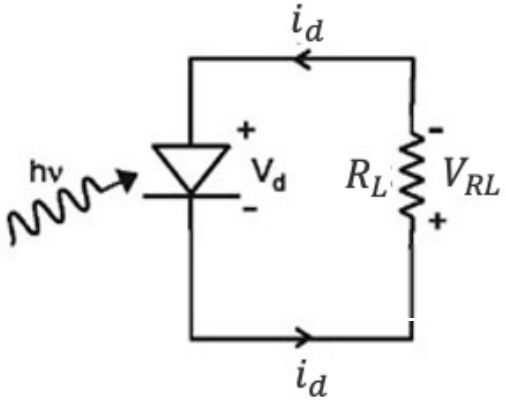


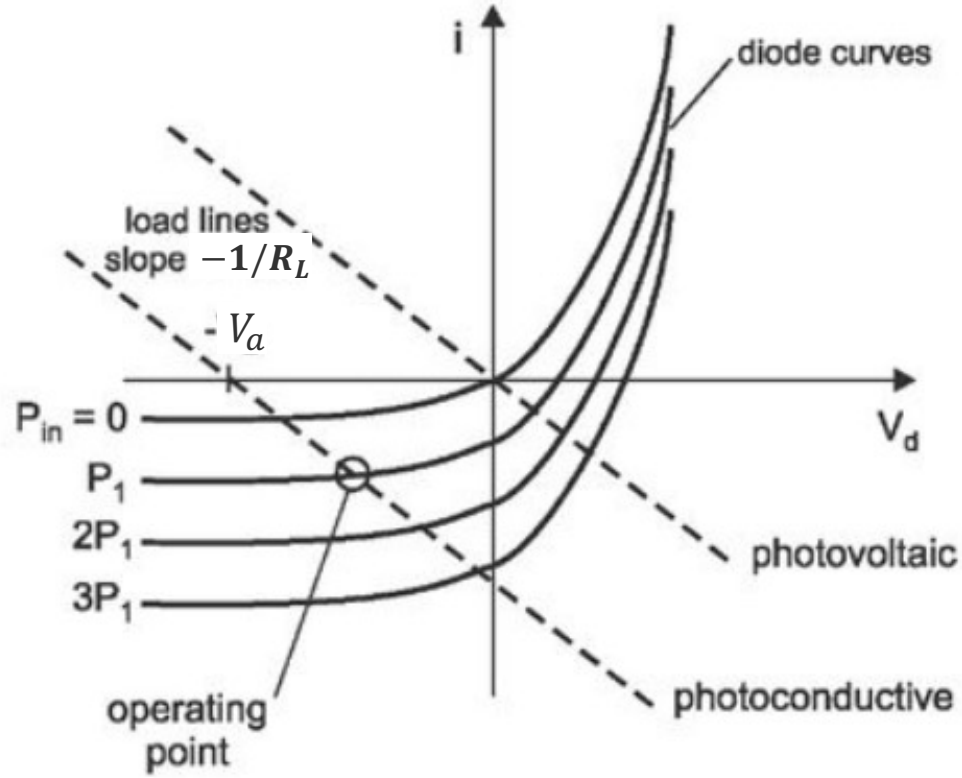
Photodiode Modes of Operation

Photoconductive Mode

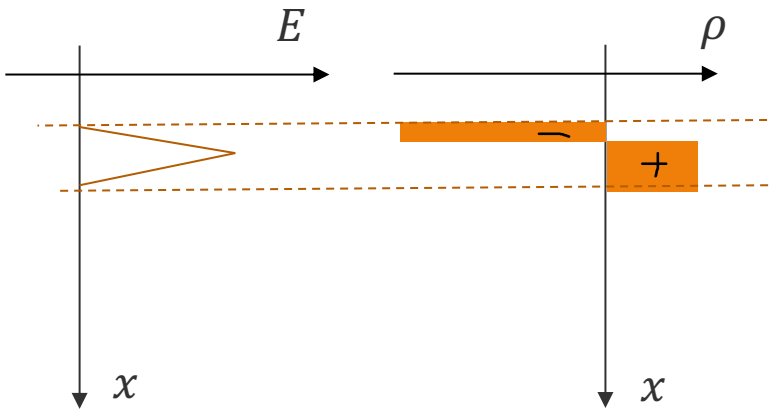


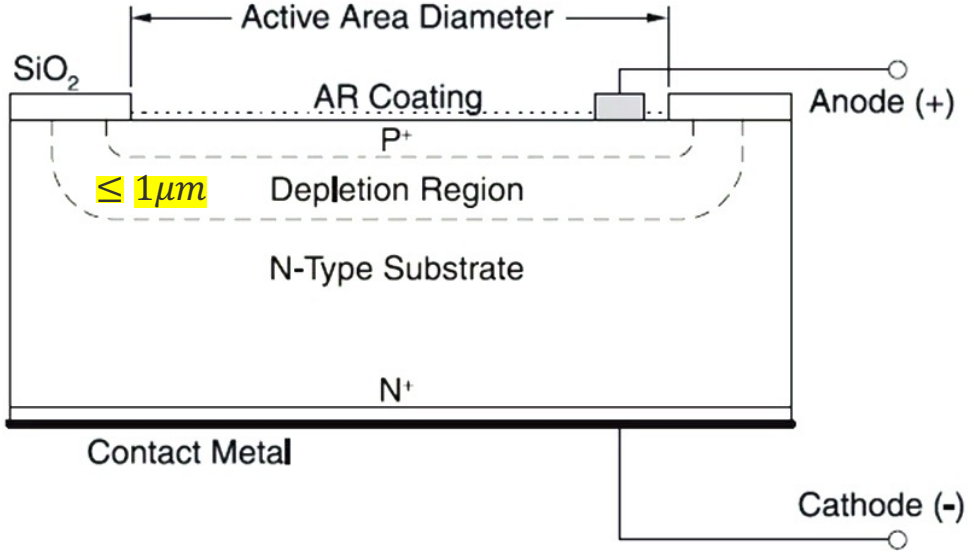
Photovoltaic Mode





Basic PN Photodiode Design

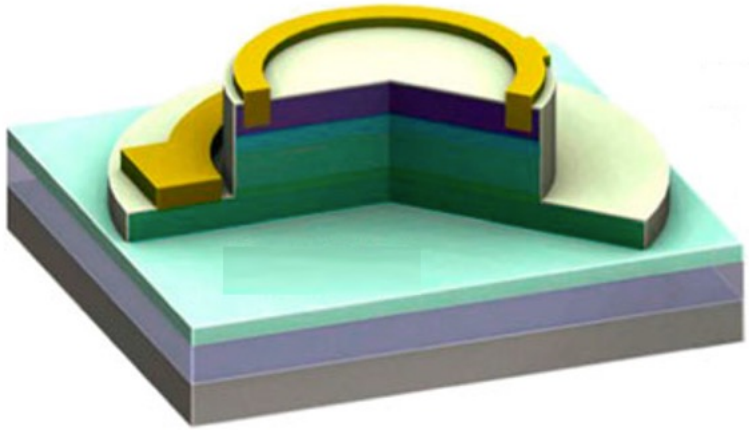




$$C_j = \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon(V_{bi}+V_A)}{q}\frac{(N_A+N_D)}{N_A N_D}}}$$

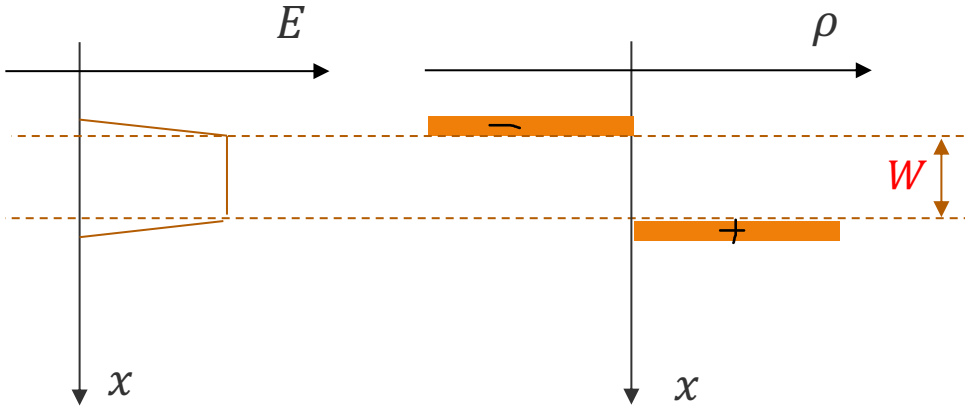
$$C_j \approx A\sqrt{\frac{q\varepsilon N_D}{2(V_{bi}+V_A)}}$$

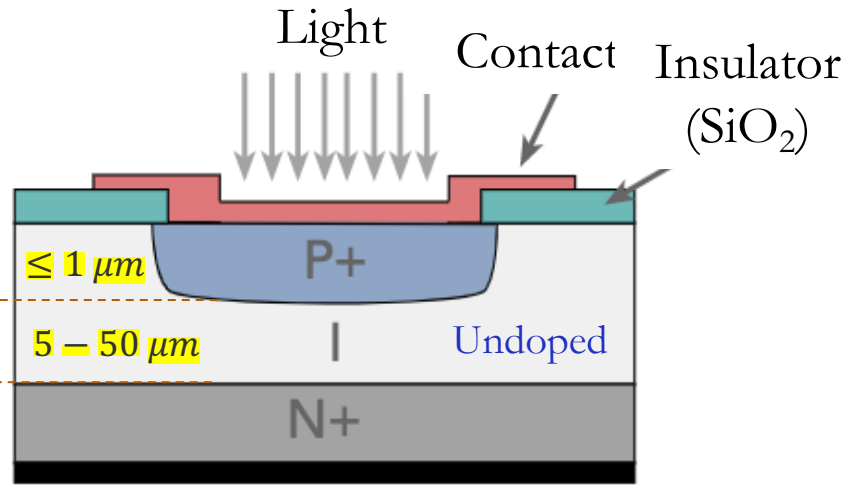
$$N_A \gg N_D$$



The PiN Photodiode

This structure improves absorption, speed and efficiency.





$$E \approx \frac{V_A}{W} \qquad C_j \approx \frac{\varepsilon A}{W} \sim pF$$

Independent of V_A

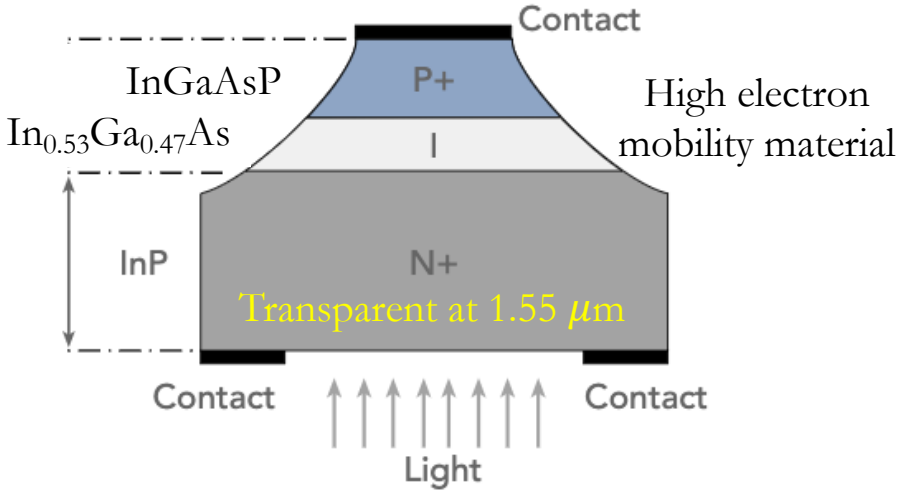
$$RC \sim 50\ pF$$

Transit time can be a problem if W is too large!

$$t = \frac{W}{v_e}$$

Increase V_A and hence v_e.
Typically 1 ns.

An excellent design for optical fiber communications.



Examples

(1) A Si PIN with L=20 μm, V_{bias}=100V, λ=900 nm (pulsed). Discuss operation.

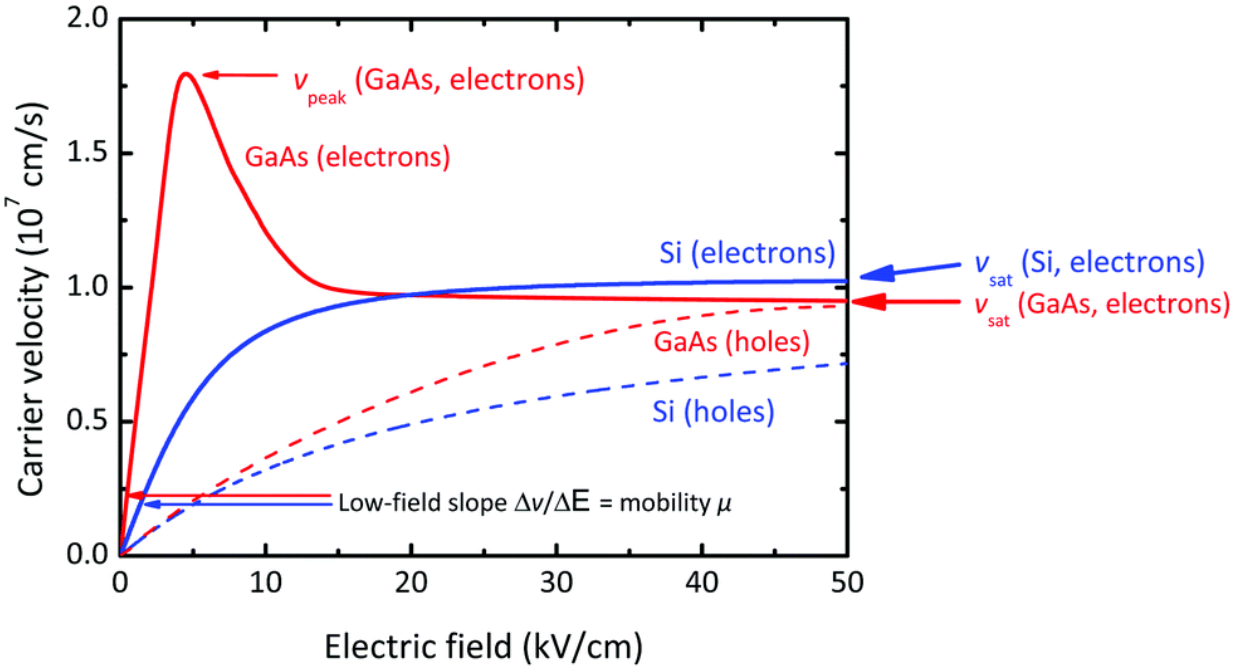
At 900 nm, $\alpha = 3\times10^4\ m^{-1}$

$$E = \frac{V_{bias}}{L} = 5\times10^6\ V/m$$

$$v_e = 10^5\ \ m/s$$

$$v_h = 7\times10^4\ \ m/s$$

$$t_h = \frac{L}{v_h} = 0.286\ ns$$



To improve the speed, decrease L, but that will hurt efficiency and hence responsivity. Find a trade-off.

Examples

(2) A Si PIN with diameter 4 cm is illuminated with a 700 nm light of intensity I=0.1 mW/m^2. The optical current produced was measured to be i_p=56.6 nA. Find efficiency and responsivity.

$$P_o = Area \times Intensity = A\,I = \pi(0.2\,mm)^2(0.1\,m)$$

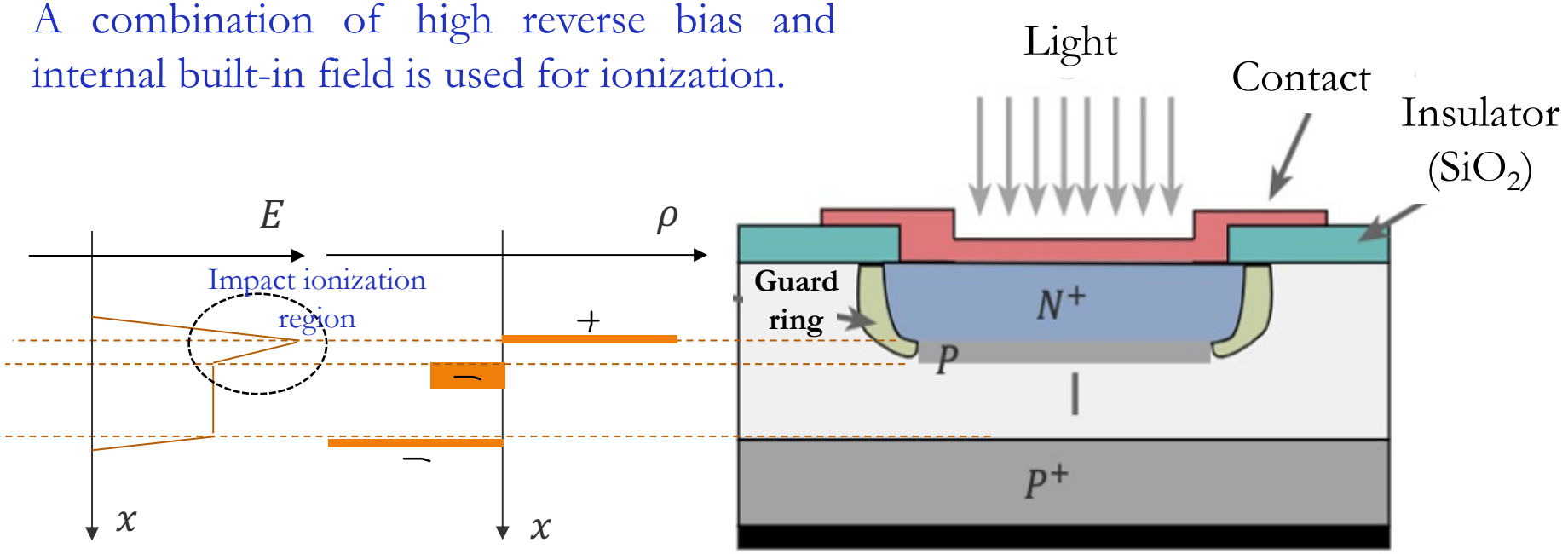
$$P_o = 1.26\times10^{-7}\,W$$

$$R = \frac{i_p}{P_o} = 0.45\,A/W$$

$$\eta = R\frac{hc}{e\lambda} = 80\%$$

The Avalanche Photodiode (APD)

A combination of high reverse bias and internal built-in field is used for ionization.



No hole ionization (lower noise and faster operation).

Ionization (Gain) factor

Si ~ 100

Ge ~ 10

AlGaAs ~ 10

Temperature-dependent

$$G \approx \frac{1}{1-(V_A/V_{br})^n}$$

$$i_p = G i_{po}$$

$$R = G\frac{i_{po}}{P_o}$$

Advantages

Avoid breakdown, high sensitivity, no need for current amplifiers, high speed (0.1 n typical).

Examples

(1) An InGaAs APD has 60% efficiency at 1550 nm with no impact ionization. Find the photocurrent and responsivity of the device if it was biased for G=12, with input power of 20 nW.

$$R_o = \frac{\eta_o q e \lambda}{hc} = 0.75\ A/W$$

$$i_{po} = R_o\ P_o = 15\ nA$$

$$i_p = G\ i_{po} = 0.18\ \mu A$$

$$R = G\ R_o = 9.0\ A/W$$

Examples

(2) A Si APD has 70% efficiency at 830 nm with no impact ionization. The photodiode is then biased for G=100. Find the photocurrent with input power of 10 nW.

$$R_o = \frac{\eta_o q e \lambda}{hc} = 0.47\ A/W$$

$$i_{po} = R_o\ P_o = 4.7\ nA$$

$$i_p = G\ i_{po} = 0.47\ \mu A$$

Q & A

Send me an email if you have any questions.